LAB ACTIVITY 3 – PART 1: THE DATA FOUNDATION

Topic: 3.1.1 JavaScript Variables, Data Types, and Operators

Duration: 45–60 Minutes

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1. Learning Outcomes

By the end of this session, you should be able to:

1. **Declare** variables using let and const appropriately.
2. **Identify** different JavaScript data types (String, Number, Boolean, Array).
3. **Perform** calculations using Arithmetic and Assignment operators.
4. **Display** data using console.log() and alert().

2. Part A: Theory (The Building Blocks)

Concept	Description	Example
Variables	Containers for storing data values.	let price = 10;
Strings	Textual data wrapped in quotes.	"Hello World"
Numbers	Numerical data (integers or decimals).	25 or 3.14
Booleans	Logical values: True or False.	isLoggedIn = true;
Operators	Symbols used to perform operations.	+, -, *, /, %

3. Part B: Practical Step-by-Step

Step 1: Variable Declaration & Setup

Goal: Set up the file structure and define basic variables.

1. Create a folder named **RegNo_Lab3_Part1**.
2. Create an index.html file.
3. Inside the <body>, insert the Header and the first Lab Card:

HTML

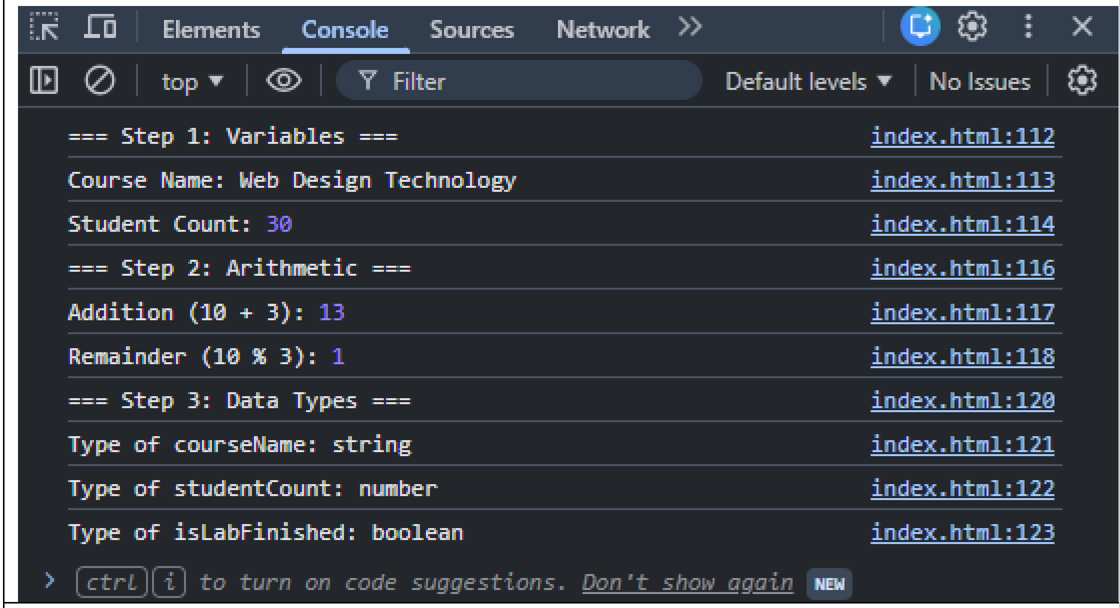
```
<header>
    <h1>Lab 3.1.1: JavaScript Fundamentals</h1>
    <p>Variables, Data Types, and Operators</p>
</header>

<section class="lab-card">
    <h2>Steps 1 - 3: Logic & Console</h2>
    <div class="instruction">
        Open the <strong>Browser Console</strong> (F12 > Console) to see
the output for variables and <code>typeof</code> checks.
    </div>
    <button onclick="runConsoleTasks()">Run Console Tasks</button>
    <p class="note">*This logs Course Name, Student Count, and Data
Types.</p>
</section>
```

4. Inside the <script> tag, define the first function covering **Steps 1, 2, and 3**:

JavaScript

```
/* ==========================================
   STEPS 1, 2 & 3: Variables & Operators
   ========================================== */
function runConsoleTasks() {
    // Variable Declaration
    const courseName = "Web Design Technology";
    let studentCount = 30;
    let isLabFinished = false;

    console.log("--- Step 1: Variables ---");
    console.log("Course:", courseName);
    console.log("Initial Students:", studentCount);

    // Arithmetic
    let x = 10;
    let y = 3;
    console.log("--- Step 2: Arithmetic ---");
    console.log("Addition (10+3):", x + y);
    console.log("Remainder (10%3):", x % y);

    // Type Identification
    console.log("--- Step 3: Data Types ---");
    console.log("Type of courseName:", typeof courseName);
    console.log("Type of studentCount:", typeof studentCount);
    console.log("Type of isLabFinished:", typeof isLabFinished);

    alert("Check your browser console (F12) to see the results!");
}
```

4. Capture a screenshot of the output from the previous steps. **Observe** the results and **explain** your understanding of the process.

Screenshot of the output

Explanation The console output shows the declared variables, arithmetic operations, and data types using the typeof operator. The console.log() function is used to display and verify the output.

Step 4: Simple UI Interaction (Prompt & Alert)

Goal: Interact with the user using built-in browser dialog boxes.

1. Add this new section to your HTML <body>:

HTML

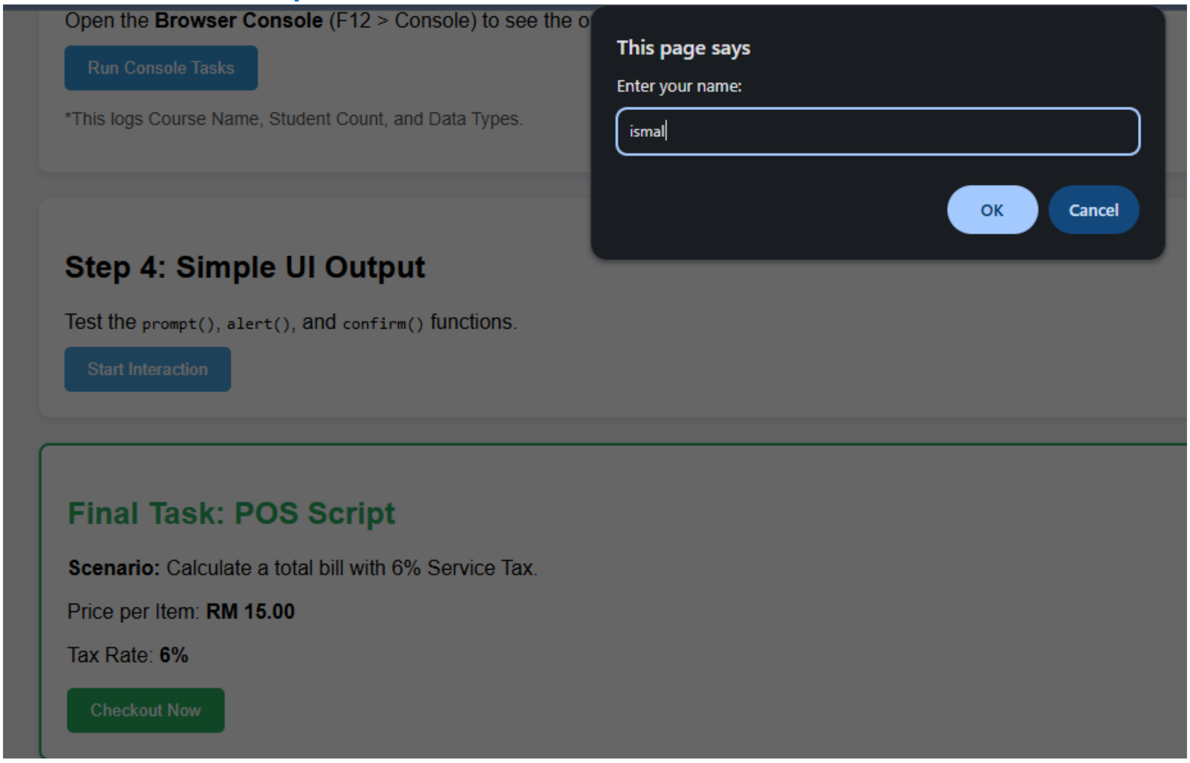
```html
<section class="lab-card">
    <h2>Step 4: Simple UI Output</h2>
    <div class="instruction">
        Test the <code>prompt()</code>, <code>alert()</code>, and
<code>confirm()</code> functions.
    </div>
    <button onclick="runInteraction()">Start Interaction</button>
</section>
```

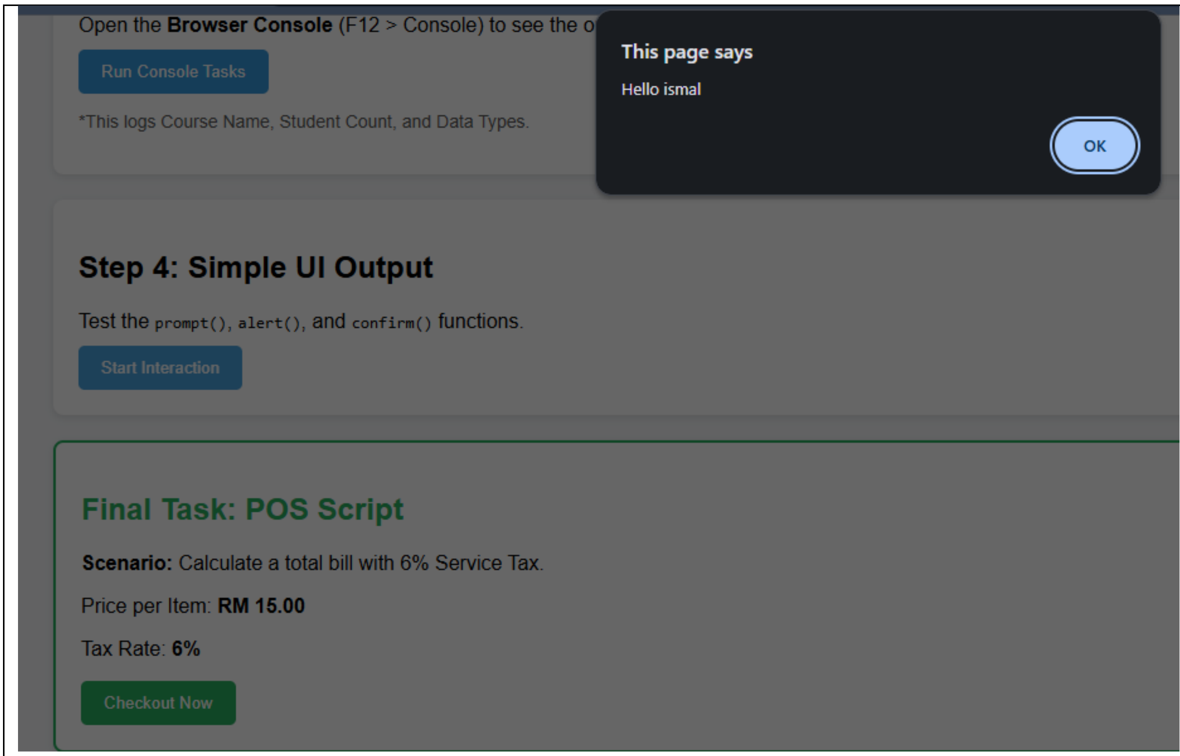
2. Add the function to your <script> tag (below runConsoleTasks):

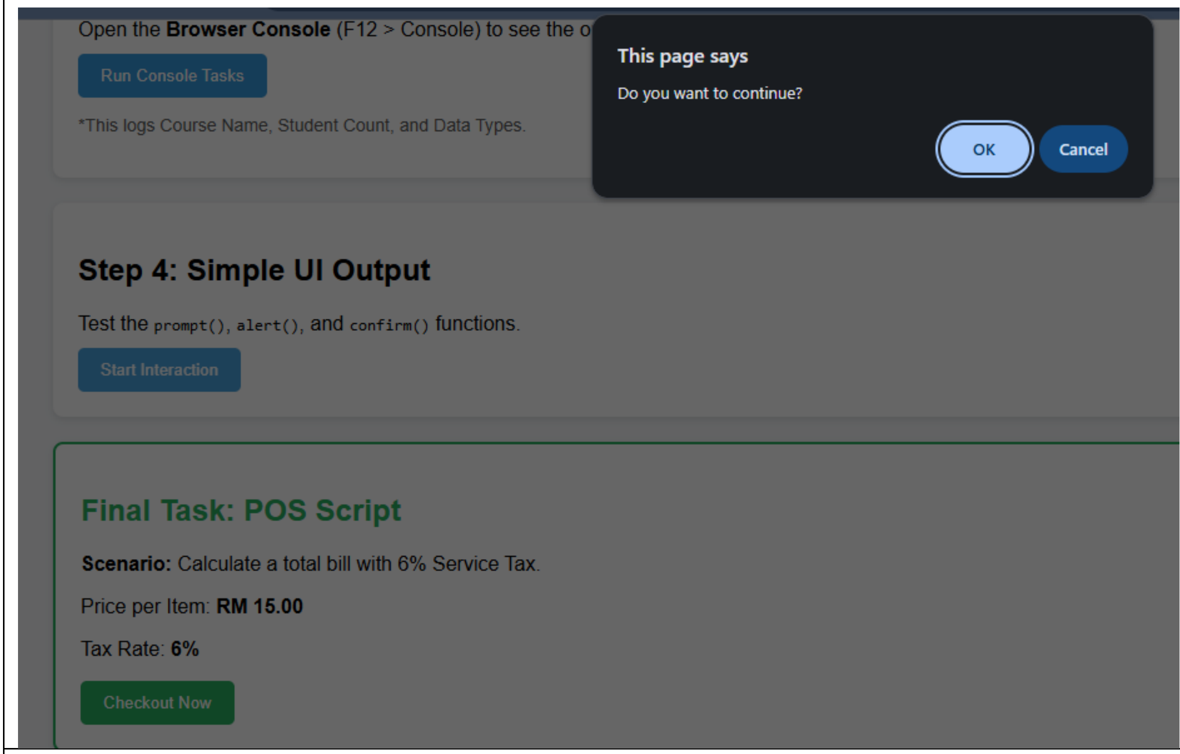
JavaScript

```
/* ==========================================
   STEP 4: Interaction
   ========================================== */
function runInteraction() {
    let username = prompt("What is your name?");
    if (username) {
        alert("Welcome to JS Lab, " + username + "!");

        let proceed = confirm("Do you want to start the lab?");
        console.log("User wants to proceed:", proceed);
    }
}
```

Screenshot of the output







Explanation

The prompt() function is used to get user input, alert() displays a message to the user, and confirm() returns a boolean value based on the user's response.

4. Final Practical Task: The POS Script

Scenario: Construct a "Point-of-Sale" script to calculate a total bill including a 6% service tax.

1. **Add the HTML UI (The Trigger):**

HTML

```html
<section class="lab-card" style="border: 2px solid #27ae60; margin-top: 20px; padding: 15px; border-radius: 8px;">
    <h2 style="color: #27ae60;">Final Task: POS Script</h2>
    <div class="instruction">
        <strong>Scenario:</strong> Calculate a total bill with 6% Service Tax.
    </div>
    <p>Price per Item: <strong>RM 15.00</strong></p>
    <p>Tax Rate: <strong>6%</strong></p>

    <button onclick="runPOS()" style="background-color: #27ae60; color: white; padding: 10px 20px; border: none; border-radius: 5px; cursor: pointer;">
        Checkout Now
    </button>
</section>
```

2. **Add the JavaScript Logic:**

JavaScript

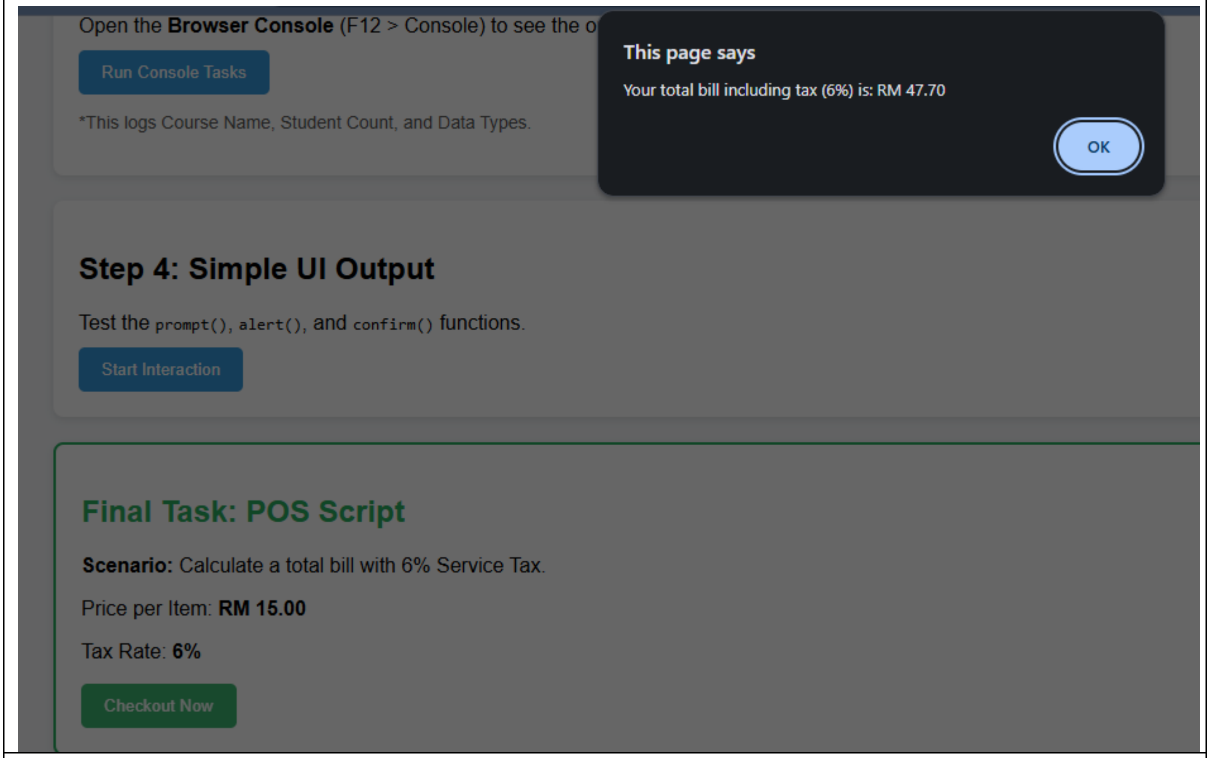
```
/* ==========================================
   FINAL TASK: Point-of-Sale (POS)
   ========================================== */
function runPOS() {
    // Constants
    const pricePerItem = 15;
    const taxRate = 0.06;

    // Input
    let quantity = prompt("How many items do you want to buy?");

    // Validation & Calculation
    if (quantity !== null && quantity !== "") {
        let qtyNumber = parseInt(quantity);

        if (!isNaN(qtyNumber) && qtyNumber > 0) {
            let subtotal = pricePerItem * qtyNumber;
            let totalTax = subtotal * taxRate;
            let finalBill = subtotal + totalTax;

            // Output with 2 decimal places
            alert("Order Summary:\n" +
                  "------------------\n" +
                  "Quantity: " + qtyNumber + "\n" +
                  "Subtotal: RM " + subtotal.toFixed(2) + "\n" +
                  "Service Tax (6%): RM " + totalTax.toFixed(2) + "\n\n" +
                  "Your total bill is: RM " + finalBill.toFixed(2));
        } else {
            alert("Invalid input! Please enter a valid number.");
        }
    }
}
```

What happens when you run this?

1. **Prompt:** Input box pops up. Type 3.
2. **Calculation:** * Subtotal:

   $$3 \times 15 = 45$$
   - Tax:
     $$45 \times 0.06 = 2.70$$
   - Total:
     $$45 + 2.70 = 47.70$$

3. **Output:** An alert box appears: **"Your total bill including tax (6%) is: RM 47.70"**.

Screenshot of the output

Explanation **The POS script calculates the total price based on the quantity entered by the user. A 6% service tax is added to the subtotal, and the final amount is displayed using toFixed(2) for proper currency formatting.**

5. Quick Check Questions

Part A

Question 1

Which of the following correctly describes the difference between the = and + operators in JavaScript?

- A. = is used for addition, while + is used to assign values to variables.
- B. = is the Assignment Operator (stores a value), while + is the Arithmetic or Concatenation Operator.
- C. Both are Arithmetic Operators used for mathematical calculations only.
- D. = compares two values, while + creates a new variable.

Question 2

What is the result of the expression "5" * 2 in JavaScript, and why?

- A. 55, because JavaScript repeats the string.
- B. Error, because you cannot multiply a string with a number.
- C. 10, because JavaScript performs "Type Coercion" to convert the string to a number.
- D. NaN (Not a Number), because the operation is mathematically impossible.

Question 3

In a Point-of-Sale (POS) system, why is it recommended to use const instead of let for a variable like taxRate = 0.06?

- A. const allows the tax rate to be changed easily later in the script.
- B. const makes the variable run faster in the browser memory.
- C. const prevents accidental reassignment, ensuring the fixed tax value remains consistent.
- D. const is the only keyword that supports decimal (float) numbers.

Part B

1. Which keyword is used to declare a variable whose value is intended to remain constant throughout the program?

A. let

B. var

C. const

D. static

2. What is the output of the following code? console.log(typeof "2026");

A. number

B. string

C. boolean

D. undefined

3. Which operator is used to find the remainder of a division between two numbers?

A. /

B. *

C. &

D. %

4. How do you display a message box to the user that requires them to click "OK" or "Cancel"?

A. alert()

B. prompt()

C. confirm()

D. console.log()

5. Which of the following is a valid Boolean value in JavaScript?

A. "true"

B. Yes

C. false

D. 1

6. What will be the result of the following operation: let x = "5" + 2;?

A. 7

B. 52

C. NaN

D. Error

7. Which method is used to convert a string into a whole number (integer)?

A. parseFloat()

B. toString()

C. parseInt()

D. toNumber()

8. In JavaScript, which operator is used for "Strict Equality" (checking both value and data type)?

A. =

B. ==

C. ===

D. !==

9. What is the purpose of the prompt() function?

A. To display a warning message.

B. To print data to the browser console.

C. To allow the user to input text data.

D. To permanently delete a variable.

10. If let a = 10; and let b = 3;, what is the result of a ** b (Exponentiation)?

A. 13

B. 30

C. 100

D. 1000

💡 JavaScript Quick Tips for Students

Step 1: Variables (`const` vs `let`)

- **const (Constant):** Gunakan untuk nilai yang tidak akan berubah. Contohnya: Nama kursus atau Kadar Cukai. Jika anda cuba ubah nilainya, JavaScript akan bagi error.
- **let:** Gunakan untuk nilai yang boleh berubah-ubah. Contohnya: studentCount (bilangan pelajar boleh bertambah atau berkurang).
- **Boolean:** Hanya ada dua nilai: true atau false. Sangat berguna untuk buat keputusan (Logic).

Step 2: Arithmetic Operators

- **+, -, *, /:** Berfungsi seperti kalkulator biasa.
- **% (Modulus):** Ini bukan peratus! Ia mencari **baki** pembahagian.
    - Contoh: 10 % 3 = 1 (kerana $3 \times 3 = 9$, bakinya adalah 1).
    - Kegunaan: Selalu digunakan untuk semak nombor genap/ganjil.

Step 3: The `typeof` Operator

- JavaScript adalah bahasa *dynamically typed*. Maksudnya, kita tak perlu beritahu komputer jika pemboleh ubah itu teks atau nombor, JavaScript akan "teka" sendiri.
- Gunakan typeof untuk **debug** kod anda jika pengiraan matematik tidak menjadi (mungkin anda tertambah Teks dengan Nombor!).

Step 4: Interaction (`prompt`, `alert`, `confirm`)

- **prompt()**: Cara paling mudah untuk dapatkan input dari pengguna. **Penting:** Input dari prompt sentiasa dalam bentuk **String (Teks)**.
- **alert()**: Untuk beri informasi sehala kepada pengguna.
- **confirm()**: Menghasilkan nilai true (jika tekan OK) atau false (jika tekan Cancel).

Final Step: POS Logic (The "Pro" Tips)

- **parseInt()**: Oleh kerana prompt() memberikan teks, kita perlu "paksa" ia menjadi nombor bulat menggunakan parseInt() supaya kita boleh buat pengiraan matematik.
- **isNaN() (Is Not a Number):** Fungsi keselamatan. Ia menyemak sama ada input pengguna itu nombor sah atau bukan (contoh: jika user taip "ABC", isNaN akan jadi true).
- **.toFixed(2)**: Sangat penting untuk aplikasi kewangan. Ia memastikan nombor kita sentiasa ada 2 tempat perpuluhan (RM 10.00, bukan RM 10).

Lab 3.1.1: JavaScript Fundamentals Quiz

1. Which keyword should you use to declare a variable that will NOT change throughout the script (e.g., a Tax Rate)?

A) `let`

B) `var`

C) `const`

D) `fixed`

2. What is the result of the expression `10 % 4` in JavaScript?

A) 2.5

B) 2

C) 4

D) 0

3. If you use `let input = prompt("Enter a number")` and the user types `5`, what is the data type of `input`?

A) Number

B) Boolean

C) Null

D) String

4. What is the purpose of the `parseInt()` function in the POS script?

A) To round a number to the nearest decimal.

B) To convert a string (text) into an integer (number).

C) To calculate the 6% service tax.

D) To display a message on the screen.

5. Which method is used to format a number to exactly two decimal places (e.g., converting 15 to 15.00)?

A) `toTwo()`

B) `math.round(2)`

C) `toFixed(2)`

D) `formatCurrency()`

Answer Key:

1. **C** (const is for constants)
2. **B** (10 divided by 4 is 2 with a **remainder of 2**)
3. **D** (All prompt inputs are returned as **strings** by default)
4. **B** (It ensures the "text" from the prompt can be used for math)
5. **C** (toFixed(2) handles the decimal formatting)